The documentation and process conversion measures necessary to comply with this revision shall be completed by 5 November 2000.

INCH-POUND

MIL-PRF-19500/291J
5 August 2000
SUPERSEDING
MIL-PRF-19500/291H
4 January 2000

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING,
TYPES 2N2906A, 2N2906AL, 2N2907A, 2N2907AL, 2N2906AUA
2N2907AUA, 2N2906AUB, AND 2N2907AUB
JAN, JANTX, JANTXV, JANJ, JANS, JANHC, AND JANKC

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, silicon, switching transistors. Five levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500 and two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (similar to a T0-18), figures 2 and 3 (surface mount case outlines UA and UB), and figures 4, 5, 6 and 7 (JANHC and JANKC).

1.3 Maximum ratings.

Types	P_T T_A = +25°C	I_C	V_{CBO}	V_{CEO}	V_{EBO}	$R_{\theta JA}$	T_J and T_{STG}
	W	mA dc	V dc	V dc	V dc	°C/W	°C
2N2906A, L, 2N2907A, L	0.5 (1)	600	60	60	5	325	-65 to +200
2N2906AUA, 2N2907AUA	0.65 (2)	600	60	60	5	210	-65 to +200
2N2906AUB, 2N2907AUB	0.5 (1)	600	60	60	5	325	-65 to +200

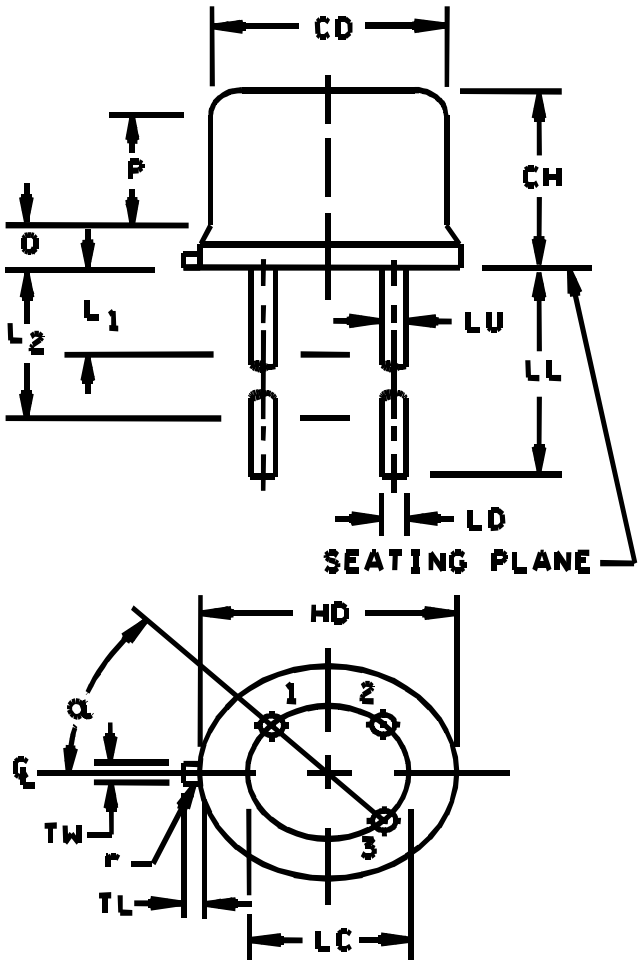
(1) Derate linearly 3.08 mW/°C above T_A = +37.5°C.
(2) Derate linearly 4.76 mW/°C above T_A = 63.5°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961
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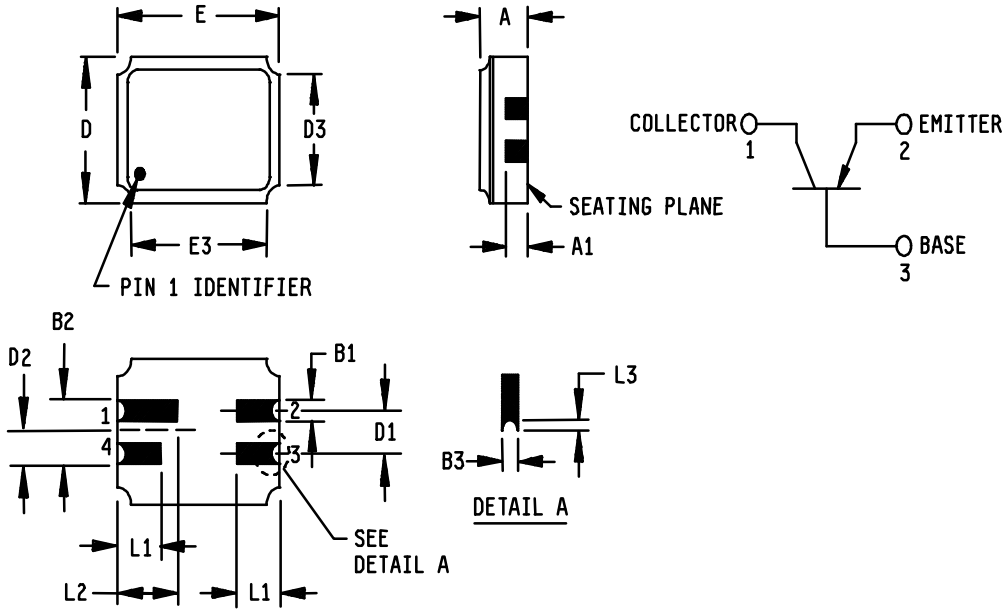
Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CD	0.178	0.195	4.52	4.95	
CH	0.170	0.210	0.8255	0.8509	
HD	0.209	0.230	5.31	5.84	
LC	100 TP		2.54 TP		6
LD	0.016	0.021	0.41	0.53	7,8
LL	0.500	0.750	12.70	19.05	7,8,13
LU	0.016	0.019	0.41	0.48	7,8
L_1	---	0.050	---	1.27	7,8
L_2	0.250	---	6.35	---	7,8
Q	---	0.030	----	0.76	5
TL	0.028	0.048	0.71	1.22	3,4
TW	0.036	0.046	0.91	1.17	3
r	---	0.010	---	0.25	10
α	45° TP		45° TP		6
P	0.100		2.54		



NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TL shall be held for a minimum length of 0.011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane 0.054 +0.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within 0.007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ⌀x symbology.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.
13. For L suffix devices, dimension LL = 1.5 inches (38.10 mm) min. and 1.75 inches (44.45 mm) max.

FIGURE 1. Physical dimensions (similar to TO-18).

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	0.061	0.075	1.55	1.90	3
A_1	0.029	0.041	0.74	1.04	
B_1	0.022	0.028	0.56	0.71	
B_2	0.075 REF		1.91 REF		
B_3	0.006	0.022	0.15	0.56	5
D	0.145	0.155	3.68	3.93	
D_1	0.045	0.055	1.14	1.39	

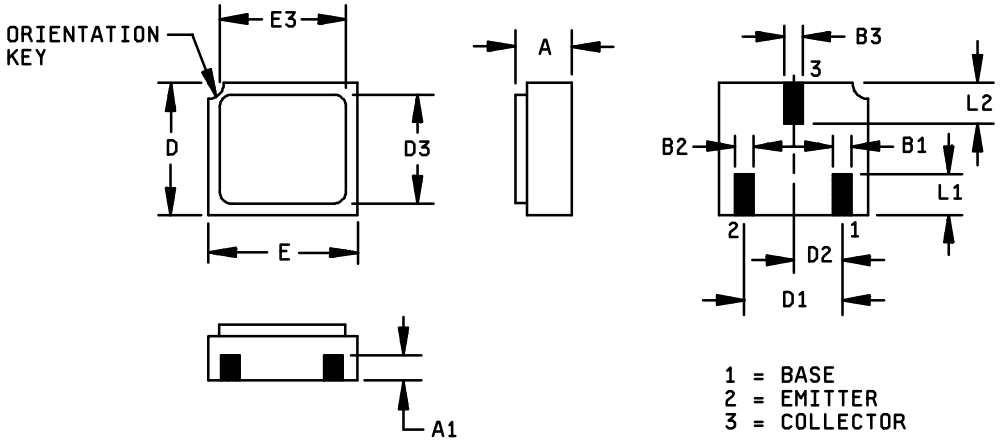
Ltr	Dimensions				Notes
	Inches		Millimeter		
	Min	Max	Min	Max	
D_2	0.0375 BSC		0.952 BSC		
D_3	---	0.155	---	3.93	
E	0.215	0.225	5.46	5.71	
E_3	--	0.225	--	5.71	
L_1	0.032	0.048	0.81	1.22	
L_2	0.072	0.088	1.83	2.23	
L_3	0.003	0.007	0.08	0.18	5

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension "A" controls the overall package thickness. When a window lid is used, dimension "A" must increase by a minimum of 0.010 inch (0.254 mm) and a maximum of 0.040 inch (1.020 mm).
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer's option, from that shown on the drawing.
5. Dimensions "B3" minimum and "L3" minimum and the appropriately castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on bottom two layers, optional on top ceramic layer.) Dimension "B3" maximum and "L3" maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.

FIGURE 2. Physical dimensions, surface mount (UA version).

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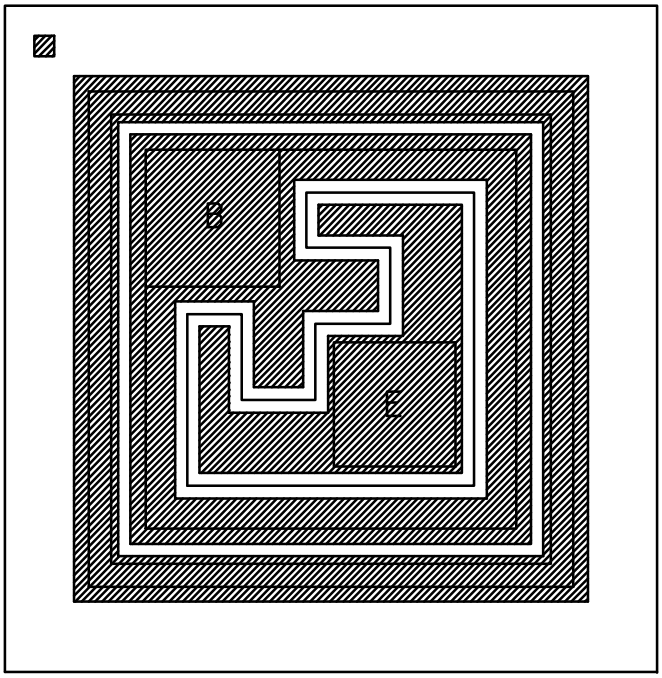


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	0.046	0.056	0.97	1.42
A1	0.017	0.035	0.43	0.89
B1	0.016	0.024	0.41	0.61
B2	0.016	0.024	0.41	0.61
B3	0.016	0.024	0.41	0.61
D	0.085	0.108	2.41	2.74
D1	0.071	0.079	1.81	2.01
D2	0.035	0.039	0.89	0.99
D3	0.085	0.108	2.41	2.74
E	0.115	0.128	2.82	3.25
E3	---	0.128	---	3.25
L1	0.022	0.038	0.56	0.96
L2	0.022	0.038	0.56	0.96

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Physical dimensions, surface mount (UB version).

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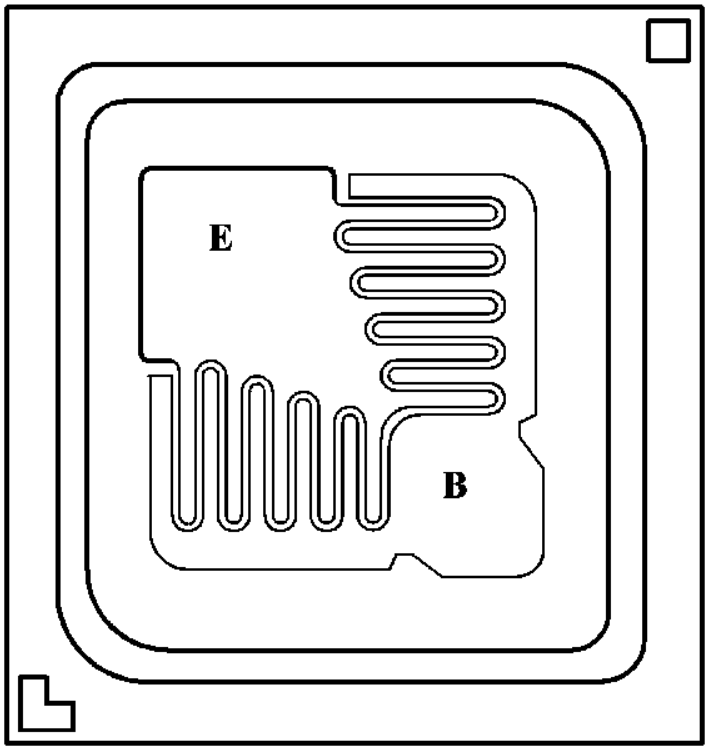
Physical characteristics:
A-version

Die size:	0.021 by 0.021 inches square
Die thickness:	0.010 ∀ 0.001 inches
Base:	0.0044 x 0.0044 inches
Emitter:	0.0041 x 0.0041 inches
Back metal:	3 K to 4.6 KÅ Gold
Top metal:	11K ± 2Å Aluminum
Back side:	Collector
Glassivation:	Si_3N_4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. JANHC and JANKC (A-version, 2N2907A only) die dimensions.

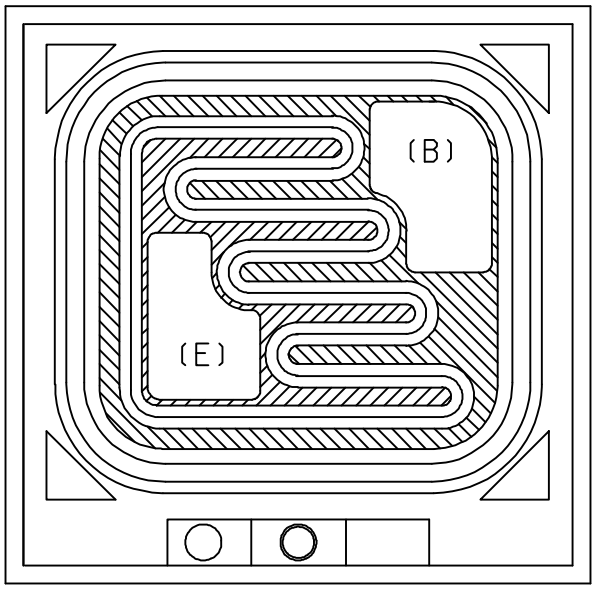
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1. Chip size	23 x 23 mils ± 2 mils.
2. Chip thickness	10 ± 1.5 mils.
3. Top metal	Aluminum 15,000 Å minimum, 18,000 Å nominal.
4. Back metal	A. Al/Ti/Ni/Ag 15kÅ/5kÅ/10kÅ/10kÅ B. Gold 2.5 kÅ minimum, 3.0 kÅ nominal. C. Eutectic Die Mount – No Metal
5. Glassivation	SI_3N_4 2k minimum, 2.2k nominal
6. Backside	Collector
7. Bonding pad	B = 4.2 x 4.2 mils, E = 4.2 x 4.2 mils

FIGURE 5. JANHC and JANKC (B-version) die dimensions.

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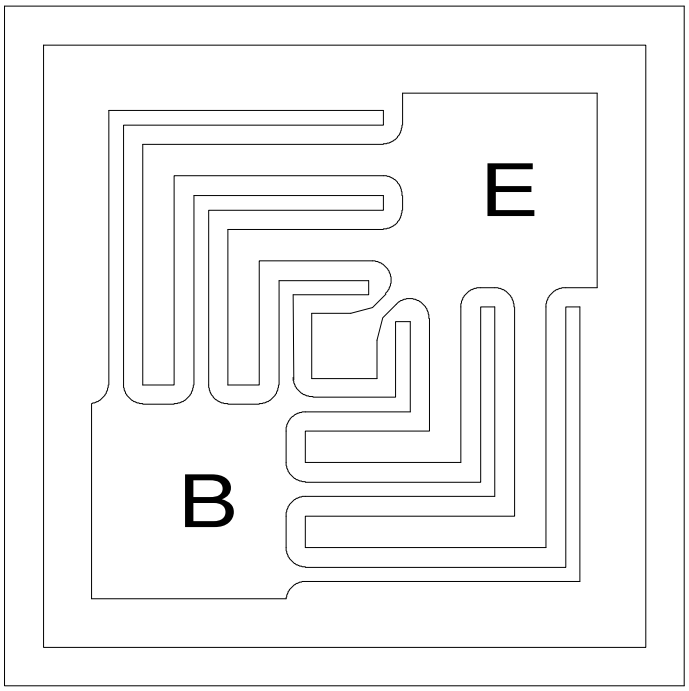


PHYSICAL CHARACTERISTICS

Die size	19.8 x 19.8 Mils square
Die thickness	10 Mils nominal thickness
Base	3.8 x 5.6 Mils
Emitter	3.8 x 5.6 Mils
Back metal	Gold, 2KÅ nominal
Back side	Collector
Glassivation	Silicon Oxide

FIGURE 6. JANHC and JANKC (C-version) die dimensions.

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Die size:	0.020 x 0.020 inches
Die thickness:	0.008 ± 0.0016 inches
Base pad:	0.004 x 0.004 inches
Emitter pad:	0.004 x 0.004 inches
Back metal	Gold, 6,500 ± 1950 Ang.
Top metal:	Aluminum, 20,000 ± 2,000 Ang.
Back side:	Collector.
Glassivation:	SiO_2, 7,500 ± 1,500 Ang.

Inches	mm	Inches	mm
0.016	0.41	0.056	1.42
0.017	0.43	0.071	1.81
0.022	0.56	0.079	2.01
0.024	0.61	0.085	2.41
0.035	0.89	0.108	2.74
0.038	0.96	0.115	2.82
0.039	0.99	0.128	3.25
0.046	0.97		

FIGURE 7. JANHC and JANKC (D-version) die dimensions.

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1.4 Primary electrical characteristics at T_A = +25°C.

h_{FE} at V_{CE} = 10 V dc										
	h_{FE1} I_C = 0.1 mA dc		h_{FE2} I_C = 1.0 mA dc		h_{FE3} I_C = 10 mA dc		h_{FE4} (1) I_C = 150 mA dc		h_{FE5} (1) I_C = 500 mA dc	
	2N2906A, L, UA,UB	2N2907A, L, UA,UB	2N2906A L, UA,UB	2N2907A L, UA,UB	2N2906A L, UA,UB	2N2907A L, UA,UB	2N2906A L, UA,UB	2N2907A L, UA,UB	2N2906A L, UA,UB	2N2907A L, UA,UB
Min	40	75	40	100	40	100	40	100	40	50
Max			175	450			120	300		

Types	Limit	$\|h_{fe}\|$ f = 100 MHz V_{CE} = 20 V dc I_C = 20 mA dc	C_{obo} 100 kHz ≤ f ≤ 1 MHz V_{CB} = 10 V dc I_E = 0	Switching (saturated) t_{on} See figure 7	Switching (saturated) t_{off} See figure 8
			pF	ns	Ns
2N2906A, 2N2907A L, UA, UB	Min	2.0			
	Max		8	45	300

Types	Limits	$V_{CE(sat)1}$ (1) I_C = 150 mA dc I_B = 15 mA dc	$V_{CE(sat)2}$ (1) I_C = 500 mA dc I_B = 50 mA dc	$V_{BE(sat)1}$ (1) I_C = 150 mA dc I_B = 15 mA dc	$V_{BE(sat)2}$ (1) I_C = 500 mA dc I_B = 50 mA dc
		V dc	V dc	V dc	V dc
2N2906A, 2N2907A L, UA, UB	Min				
	Max	0.4	1.6	1.3	2.6

(1) Pulsed see 4.5.1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1, 2, 3, 4, 5 and 6 herein. Epoxy die attach may be used when a moisture monitor plan has been submitted and approved by the qualifying activity.

3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturers symbol or logo. The prefixes JAN, JANTX, JANTXV, JANJ and JANS can be abbreviated as J, JX, JV, JJ and JS respectively. The "2N" prefix and the "AUB" suffix can also be omitted.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in Table I, group A herein.

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4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.2.1 JANJ qualification. For JANJ level, 3.3.1 through 3.3.1.3 of MIL-PRF-19500 shall apply, except as modified herein. Supplier imposed requirements as well as alternate screens, procedures, and/or controls shall be documented in the QM plan and must be submitted to the Qualifying Activity for approval. When alternate screens procedures, and/or process controls are used in lieu of the JANJ screens herein equivalency shall be proven and documented in the QM Plan. Radiation characterization may be submitted in the QM plan at the option of the manufacturer, however, paragraph 3.3.1.1 of MIL-PRF-19500 is not required. Lot formation and Conformance Inspection requirements for JANJ shall be those used for JANTXV devices as a minimum. Die lot controls and rework requirements shall be in accordance with MIL-PRF-19500 paragraphs 3.13 and D.3.13.2.1.

4.2.1.1 JANJ Qualification. For JANJ qualification, 4.4.2.1 herein shall be performed as required by the Qualifying Activity.

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4.3 Screening (JANTX, JANTXV, JANJ, and JANS levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS Level	JANJ Level
1b	Required	Required
2	Optional	Optional
3a 3b 3c (1)	Required Not Applicable Required (see 4.3.3)	Required Not Applicable Required (see 4.3.3)
4	Required	Optional
5	Required	Required (one pass minimum)
7a and 7b (3)	Required	Required
8	Required	Not Required
9	I_{CBO2}, h_{FE4} read and record	I_{CBO2}, h_{FE4} (Bin and cell)
10	24 hours minimum	24 hours minimum
11	I_{CBO2}; h_{FE2}; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE4} = ±15 percent	I_{CBO2}; h_{FE2}; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE4} = ±15 percent
12	See 4.3.2, 240 hours minimum	See 4.3.2, 160 hours minimum
13 (2)	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15 percent	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15 percent
14a, and 14b (3)	Optional	Optional
15	Required	Required - Attributes Data only, Film or Non-Film Techniques may be utilized
16	Required	Required

(1) Thermal impedance may be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.

(2) PDA = 5 percent for screen 13, applies to ΔI_{CBO2}, Δh_{FE4}, I_{CBO2}, and h_{FE4}. Thermal impedance ($Z_{\theta JX}$) is not required in screen 13.

(3) Hermeticity evaluation may be performed in either step 7 or step 14.

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4.3 Screening (JANTX, JANTXV, JANJ, and JANS levels only) Continued.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANTXV and JANTX Level
1b	Required (JANTXV only)
2	Optional
3a	Required
3b 3c (1)	Not Applicable Required (see 4.3.3)
4	Optional
5 and 6	Not Applicable
7a and 7b (2)	Required
8	Not Required
9	Not Applicable
10	24 hours minimum
11	I_{CBO2} , h_{FE4}
12	See 4.3.2, 80 hours minimum
13	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15 percent
14a and 14b	Optional
15 and 16	Not Required

(1) Thermal impedance may be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.
(2) Hermeticity evaluation may be performed in either step 7 or step 14.

4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100 percent probed to insure the assembled chips will meet the requirements of group A, subgroup 2.

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4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750, 4.5, Power shall be applied to the device to achieve T_J = minimum 135°C and minimum power dissapation of P_D = 75 percent P_T max as defined in 1.3.

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current ..... 5 mA.

b. I_H forward heating current ..... 200 mA (min).

c. t_H heating time ..... 25 - 30 ms.

d. t_{md} measurement delay time ..... 60 µs max.

e. V_{CE} collector-emitter voltage ..... 10 V dc minimum

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 60°C/W. (UA, UB); 67°C/W (TO-18).

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Group A inspection for JANJ shall be in accordance with JANTXV of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein. See 4.4.2.2 for JAN, JANTX, JANJ and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, JANJ and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with group A, subgroup 2 and 4.5.3 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 - 30 V dc.
B5	1027	V_{CB} = 10 - 30 V dc; T_A = +125°C ± 25°C for 96 hours with P_T adjusted according to the chosen T_A to give T_J = +275°C minimum. Optionally this test may be performed for 216 hours minimum with P_T adjusted to achieve T_J = + 225°C minimum, n = 45 devices, c = 0. In this case the ambient temperature shall be adjusted such that a minimum of 75 percent of maximum rated P_T (see 1.3) is applied to the DUT. (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.)

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4.4.2.2 Group B inspection, (JAN, JANTX, JANTXV and JANJ). Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new "assembly lot" option is exercised, the failed assembly lot shall be scrapped.

Step	Method	Condition
1	1027	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 - 30 V dc, Power shall be applied to the device to achieve T_J = +150°C minimum, and minimum power dissipation of 75 percent of max rated P_T (see 1.3 herein); n = 45, C = 0.
2	1027	The steady-state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B step 2 shall not be required more than once for any single wafer lot. n = 45, C = 0.
3	1032	High- temperature life (non operating), T_A = +200°C, t = 340 hours, n = 22, C = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, JANTXV and JANJ samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, JANTXV and JANJ) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS).and 4.4.3.2 (JAN, JANTX, JANTXV and JANJ) herein for group C testing. Electrical measurements (end points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, (not applicable for UA and UB devices).
C6	1027	Test condition B, 1,000 hours, V_{CB} = 10 - 30 V dc, Power shall be applied to the device to achieve T_J = +150°C minimum, and minimum power dissapation of 75 percent of max rated P_T (see 1.3 herein); n = 45, C = 0.

4.4.3.2 Group C inspection, table VII (JAN, JANTX, JANTXV and JANJ) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UA and UB devices.
C6		Not applicable.

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4.4.3.3 Group C sample selection. Samples for steps in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be performed for qualification or requalification only. The tests specified in table II herein must be performed to maintain qualification.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Delta requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit
		Method	Conditions		
1	Collector-base cutoff current	3036	Bias condition D, V_{CB} = 50 V dc	ΔI_{CB02} (1)	100 percent of initial value or 10 nA dc, whichever is greater.
2	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed see 4.5.1	Δh_{FE4} (1)	± 25 percent change from initial reading.

(1) Devices which exceed the group A limits for this test shall not be accepted.

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical inspection 3/	2071	n = 45 devices, c = 0				
Solderability 3/ 5/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 5/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic seal 5/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 5/		Group A, subgroup 2				
Bond strength 3/ 5/	2037	Precondition T_A = + 250°C at t = 24 hours or T_A = + 300°C at t = 2 hours n = 11 wires, c = 0				
Subgroup 2						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO1}		10	uA dc
Cutoff current, emitter to base	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO1}		10	uA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 50 V dc	I_{CES}		50	nA dc
Collector to base cutoff Current	3036	Bias condition D; V_{CE} = 50 V dc	I_{CBO2}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO2}		50	nA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE1}			
Forward-current transfer ratio				40		
2N2906A, L, UA, UB				75		
2N2907A, L, UA, UB						
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}			
2N2906A, L, UA, UB				40	175	
2N2907A, L, UA, UB				100	450	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE3}			
2N2906A, L, UA, UB				40		
2N2907A, L, UA, UB				100		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE4}			
2N2906A, L, UA, UB				40	120	
2N2907A, L, UA, UB				100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE5}			
2N2906A, L, UA, UB				40		
2N2907A, L, UA, UB				50		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.4	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.6	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.6	1.3	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.6	

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO3}		10	µA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE6}			
2N2906A, L, UA, UB 2N2907A, L, UA, UB				20 50		
Subgroup 4						
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}			
2N2906A, L, UA, UB 2N2907A, L, UA, UB				40 100		
Magnitude of small- signal short- circuit forward current transfer ratio	3306	V_{CE} = 20 V dc; I_C = 20 mA dc; f = 100 MHz	$\|h_{fe}\|$	2.0		
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Input capacitance (output open- circuited)	3240	V_{EB} = 2.0 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		30	pF
Saturated turn-on time		(See figure 8)	t_{on}		45	ns
Saturated turn-off time		(See figure 9)	t_{off}		300	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroups 5 and 6 Not applicable Subgroup 7 4/ Decap internal visual (design verification)	2075	n = 1 device, c = 0				

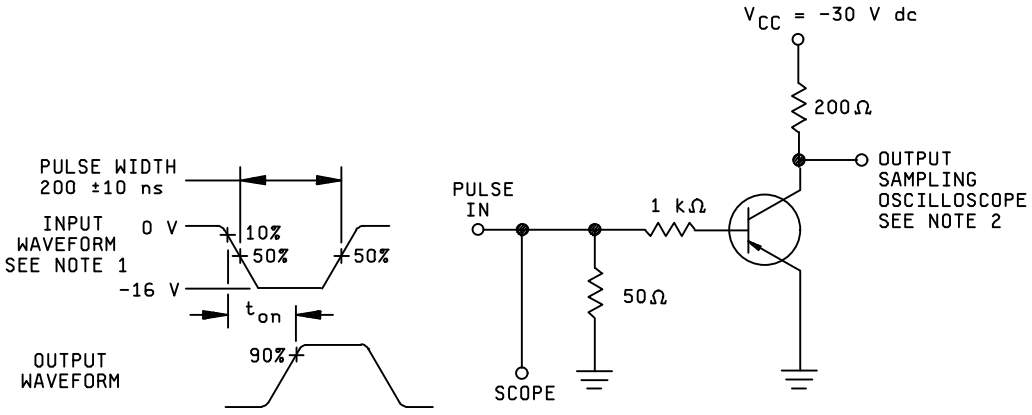
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ Not required for JANS devices.

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TABLE II. Group E inspection (all quality levels) - For qualification only

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See group A, subgroup 2 and 4.5.3 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc , 6,000 cycles.	
Electrical measurements		See group A, subgroup 2 and 4.5.3 herein.	
Subgroup 3			
Not applicable			
Subgroup 4			
Not applicable			
Subgroup 5			
Not applicable			

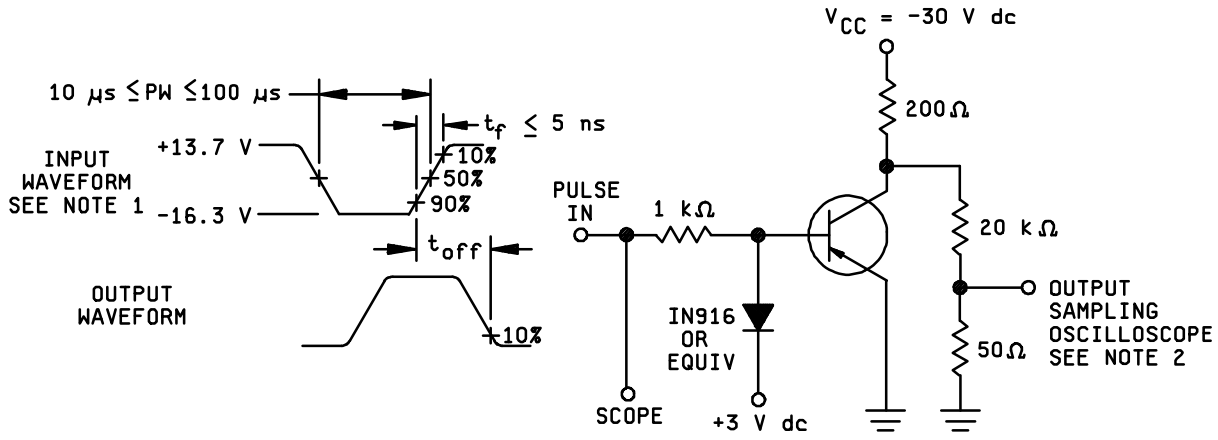
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NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 K ohms, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 8. Saturated turn-on switching time test circuit.



NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 K ohms, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 9. Saturated turn-off switching time test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation (see 2.1.1).

b. The lead finish as specified (see 3.4.1).

c. Type designation and quality assurance level.

d. Packaging requirements (see 5.1).

e. For die acquisition, the letter version must be specified (see figures 4, 5 and 6).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

6.4 Supersession information. Devices covered by this specification supersede the manufacturers' and users' Part or Identifying Number (PIN). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification. This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

6.5 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N2907A) will be identified on the QPL.

Die ordering information (1)				
PIN	Manufacturer			
	32694	43611	12969	34156
2N2906A 2N2907A	JANHCA2N2907A	JANHCB2N2906A JANHCB2N2907A	JANHCC2N2906A JANHCC2N2907A	JANHCD2N2906A JANHCD2N2907A

(1) For JANKC level, replace JANHC with JANKC.

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA – NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19

Preparing activity:
DLA - CC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/291J	2. DOCUMENT DATE 000805

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING, TYPES 2N2906A, 2N2906AL, 2N2907A, 2N2907AL, 2N2906AUA 2N2907AUA, 2N2906AUB, AND 2N2907AUB JAN, JANTX, JANTXV, JANJ, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial DSN FAX EMAIL 614-692-0510 850-0510 614-692-6939 alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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